

Lab 10: Build and Test Reusable Functions

Level: Intermediate Mapping: LO5 | K7 | A7 | A9 Concepts: functions, default arguments, named arguments, ..., set.seed

Scenario

A game-based learning team needs a reliable two-dice simulator and summary function.

Goal

Two documented functions with deterministic tests.

Files in this folder

- lab-10.R - completed guided R script
- verify.R - repeatable acceptance check
- lab-workbook.xlsx - brief, input data, controls and evidence record
- input-data.csv - portable fallback if readxl is unavailable

Detailed step-by-step

1. Define roll_dice(n=2, sides=6) with input validation.
2. Use sample() and sum() to return a roll total.
3. Define summarise_rolls(x, ...) and forward options to mean().
4. Set a random seed and run deterministic tests.
5. Record expected and actual results in the workbook.

Run and verify

1. Set the RStudio working directory to this lab folder.
2. Open lab-10.R and read the header, inputs and validation rules.
3. Run the complete script with source("lab-10.R").
4. Confirm the Console ends with LAB VERIFIED: outputs created and checks passed.
5. Run source("verify.R") and retain the verification message.
6. Open analysis_output.csv and compare the evidence with the workbook Control sheet.
7. Complete the workbook Evidence sheet and save it without changing the filename.

Acceptance criteria

- [] Invalid n or sides values stop with clear messages.
- [] Default and named argument calls both work.
- [] Tests are reproducible under the recorded seed.

Troubleshooting

- there is no package called 'readxl': the script automatically uses input-data.csv; install readxl later if Excel import is required.
- cannot open file: confirm RStudio's working directory is this exact lab folder.
- object not found: run the script from the top so imports and validations occur before analysis.
- Verification fails: inspect the first error, compare column names with Input_Data, and rerun from a clean session.

Evidence to submit

Completed Excel Evidence sheet, R script, generated output files and the final Console verification message.